

OPTICROP: SMART AGRICULTURAL PRODUCTION OPTIMIZATION ENGINE

CODE GUIDELINES AND LAYOUT RULES

1. Project Directory Structure

To maintain code clean and readable, files are structured into functional folders:

- Root: Entry points `app.py`, `train_models.py`, `generate_datasets.py`.
- `datasets/`: Raw agricultural databases and logged user history.
- `models/`: Saved pickle (`.pkl`) files (scalers, encoders, estimators).
- `templates/`: HTML view pages (`index`, `result`, `dashboard`, `weather`, `ai_advisor`).
- `static/`: CSS styling and generated performance visualizations.
- `tests/`: Unit test scripts.

2. Naming and Formatting Standards

Python modules follow PEP-8 conventions. Methods use clear names (e.g. `'load_all_models()'`, `'train_and_compare()'`), and variables include clear descriptions. Models are saved as pipelines to preserve the execution order of standardizers and encoders.